

Progress Status Summary and New Experiments

I returned back to the earlier improvement plan that I sent to you and then studied what you are achieved, the following table summarizes which experiments are done, partially done, or new additions.

Experiment	Status	Notes
Baseline (features only)	Done	98% binary, 72% multi-class
Baseline + BERT	Done	99.6% binary, 81% multi-class
Misspelling attack	Done	Needs multi-rate sweep (5%, 10%, 15%, 20%)
Cross-domain (Essay + WP)	Done	Needs per-class breakdown
Unseen model (Reuters/Claude)	Done	Needs per-class breakdown
Paraphrasing (Gemma-27b)	Done	Multi-class is very weak, it needs analysis
Round-trip translation (NLLB)	Done	Near-perfect results
Input length sensitivity	New	Short/Medium/Long bin analysis
Misspelling rate sweep	New	Degradation curve at 5%, 10%, 15%, 20%
Feature ablation study	New	Remove feature groups one at a time
Per-class confusion analysis	Partial	Confusion matrices exist; need per-class F1 tables for all experiments

In the following, I describe each new experiment or enhancement experiments separately. In total you have SIX tasks (or experiments). They are explained as follows:

Phase 1: Experiments

Dear Basel. All tasks in this phase produce **numerical results, tables, and figures**. So, **you should save all outputs in a shared folder with consistent naming**. Each experiment must record: the exact random seed, the train/test split used, and the feature set (with or without BERT).

1.1 Baseline Consolidation (Enhance Existing)

Status: Mostly done. Needs additional per-class reporting.

Deliverables:

- Confirm fixed random seed and 80/20 split across all experiments.
- Generate per-class precision, recall, and F1 for both tasks (binary and 7-class), for both configurations (features-only and features+BERT).
- Save and label all confusion matrices as high-resolution PNG files (300 DPI minimum).
- Save feature importance bar charts for both configurations.

Required output tables:

Table T1: Binary Detection Results (AI vs. Human)

Configuration	Accuracy	Precision	Recall	F1-Score
Features only	0.98	0.97	0.96	0.97

Features + BERT	0.996	...	...	...
-----------------	-------	-----	-----	-----

Table T2: Multi-class Attribution Results (7 classes)

Configuration	Accuracy	Precision	Recall	F1-Score
Features only	0.72	0.72	0.72	0.72
Features + BERT	0.81	0.81	0.81	0.81

Table T3: Per-class F1 Scores for Multi-class Attribution

Config	Human	GPT-4o	Gemma	Qwen	LLaMA	Mistral	Yi-Large
Features only	...	...	...	...	...	...	
Features + BERT	...	...	...	...	...	...	

Required figures:

- Figure F1: Confusion matrix for binary detection (features+BERT).
- Figure F2: Confusion matrix for multi-class attribution (features+BERT).
- Figure F3: Feature importance bar chart (features only, binary).
- Figure F4: Feature importance bar chart (features+BERT, binary).
- Figure F5: Feature importance bar chart (features only, multi-class).
- Figure F6: Feature importance bar chart (features+BERT, multi-class).

1.2 Misspelling Rate Sweep (New Experiment)

Status: Current results at 20% only. Need to run at multiple rates.

The current 20% misspelling rate causes catastrophic failure (19.6% accuracy). This makes the results hard to interpret. Running at multiple perturbation levels will produce a degradation curve that tells a much better story.

Deliverables:

- Run the existing misspelling attack at 5%, 10%, 15%, and 20% perturbation rates.
- Report accuracy, precision, recall, and F1 for both binary and multi-class tasks, at each rate, for both configurations (features-only and features + BERT).

Table T4: Misspelling Attack - Binary Detection Degradation

Perturbation %	Features F1	Feat+BERT F1	Features Acc	Feat+BERT Acc
0% (baseline)	0.97	...	0.98	...
5%	...	...	...	...
10%	...	...	...	...
15%	...	...	...	...
20%	...	...	...	...

Required figures:

- Figure F7: Line chart showing accuracy degradation vs. perturbation rate (binary task, both configs on same plot).
- Figure F8: Same as F7 but for multi-class attribution.

1.3 Input Length Sensitivity Analysis (New Experiment)

This experiment tests whether the model performs differently on short vs. long texts, which has practical implications for real-world deployment.

Deliverables:

- Partition the test set into three bins based on word count: Short (bottom 33rd percentile), Medium (middle 33rd), Long (top 33rd). Report the actual word-count boundaries.
- Evaluate binary and multi-class accuracy per bin, for both configurations.
- Report which features dominate in each length bin (top 5 by importance).

Table T5: Length Sensitivity - Binary Detection

Length Bin	N samples	Features Acc	Features F1	Feat+BERT F1
Short	...	...	...	...
Medium	...	...	...	...
Long	...	...	...	...

Required figures:

- Figure F9: Grouped bar chart showing F1 across length bins (binary + multi-class, both configs).

1.4 Cross-Domain Generalization (Extend Existing)

Status: Done at aggregate level. Needs per-class detail.

Deliverables:

- For Essay and WP benchmarks: report per-class precision, recall, and F1 (not just macro averages).
- Generate confusion matrices for each benchmark.

Table T6: Cross-Domain Binary Detection

Benchmark	Features Acc	Feat+BERT Acc	Features F1	Feat+BERT F1
Original test set	0.98	0.996	0.97	...
MTG Essay (GPT4All vs Human)	0.705	0.609	0.679	0.540
MTG WP (GPT4All vs Human)	0.927	0.824	0.927	0.820
Reuters/Claude (unseen model)	0.672	0.835	0.649	0.835

Required figures:

- Figure F10: Confusion matrices for Essay and WP benchmarks (features+BERT).
- Figure F11: Confusion matrix for Reuters/Claude unseen model test.

1.5 Paraphrasing and Translation Attacks (Extend Existing)

Status: Done at aggregate level. Multi-class paraphrasing results need investigation.

Deliverables:

- For paraphrasing: generate confusion matrices for both binary and multi-class tasks.
- For multi-class paraphrasing: report per-class F1 to identify which models are most confusable after paraphrasing.
- For translation: confirm results and generate confusion matrices.

Table T7: Adversarial Attack Results Summary (Binary)

Attack	Features Acc	Feat+BERT Acc	Features F1	Feat+BERT F1
None (baseline)	0.98	0.996	0.97	...
Paraphrasing (Gemma-27b)	0.802	0.889	0.832	0.857
Translation (NLLB)	0.996	1.000	0.996	1.000
Misspelling (20%)	0.196	0.390	0.138	0.430

Required figures:

- Figure F12: Confusion matrix for paraphrasing attack (multi-class, features+BERT).

1.6 Feature Ablation Study (New Experiment)

This experiment systematically removes feature groups to quantify each group's contribution. This strengthens the paper's analytical depth beyond just reporting feature importance scores.

Deliverables:

- Define feature groups: (a) Perplexity + UID, (b) Burstiness, (c) TTR + Stylometry, (d) LIWC/Empath features, (e) Semantic consistency, (f) Syntax depth, (g) BERT embeddings.
- Train XGBoost with all features minus one group at a time. Report accuracy and F1 for both tasks.

Table T8: Feature Ablation - Impact of Removing Each Feature Group

Removed Group	Binary Acc	Binary F1	Multi Acc	Multi F1
None (full model)	...	...	...	...
Perplexity + UID	...	...	...	...
Burstiness	...	...	...	...
TTR + Stylometry	...	...	...	...
LIWC/Empath	...	...	...	...
Semantic Consistency	...	...	...	...
Syntax Depth	...	...	...	...
BERT Embeddings	...	...	...	...

Required figures:

- Figure F13: Horizontal bar chart showing F1 drop when each feature group is removed (both tasks).

In summary all deliverables will be:

ID	Description	Status
T1	Binary detection results (features vs. features+BERT)	Partial
T2	Multi-class attribution results (features vs. features+BERT)	
T3	Per-class F1 for multi-class attribution	New
T4	Misspelling attack degradation at 5%, 10%, 15%, 20%	New
T5	Length sensitivity analysis (short/medium/long)	New
T6	Cross-domain generalization results	Partial
T7	Adversarial attack summary (all attacks, binary)	Partial
T8	Feature ablation study results	New

Figures (Basel generates; Prof. Sagheer selects and captions for paper):

ID	Description	Status
F1	Confusion matrix: binary detection (features+BERT)	Exists
F2	Confusion matrix: multi-class attribution (features+BERT)	Exists
F3	Feature importance: binary, features only	Exists
F4	Feature importance: binary, features+BERT	New
F5	Feature importance: multi-class, features only	Exists
F6	Feature importance: multi-class, features+BERT	New
F7	Degradation curve: misspelling vs. accuracy (binary)	New
F8	Degradation curve: misspelling vs. accuracy (multi-class)	
F9	Bar chart: length sensitivity (both tasks, both configs)	
F10	Confusion matrices: Essay and WP benchmarks	New
F11	Confusion matrix: Reuters/Claude unseen model	New
F12	Confusion matrix: paraphrasing attack (multi-class)	New
F13	Bar chart: feature ablation impact	New

How to organize the outputs?

- Please organize all outputs in the following folder structure:
 - results/tables/ - All result tables as CSV files (one per table, named T1.csv through T8.csv).
 - results/figures/ - All figures as PNG at 300 DPI (named F1.png through F13.png).
 - results/models/ - Saved model files (.pkl or .joblib) with metadata (seed, split, feature list).
 - results/logs/ - Training logs showing hyperparameters used.
- Every experiment must use the same random seed (document which one) and the same 80/20 split.
- **Please complete Task 1.1 first before proceeding to other tasks.**